llm-guided physics simulation

for non-programmer

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intro

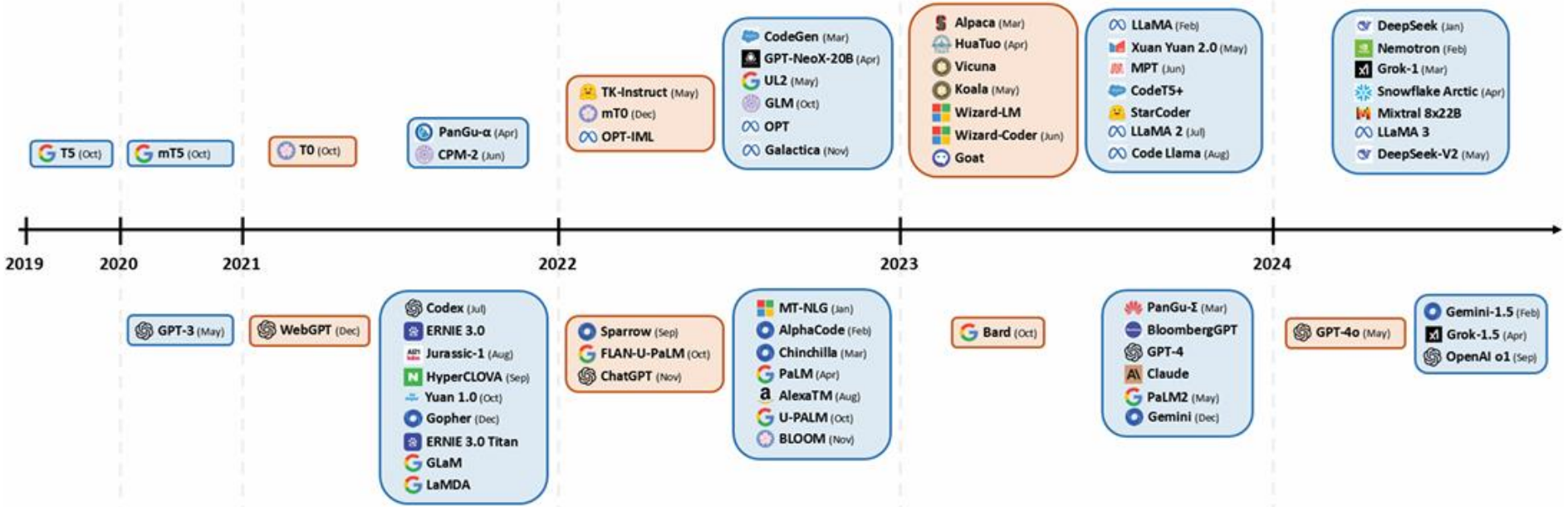
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Historical Progress of NLP

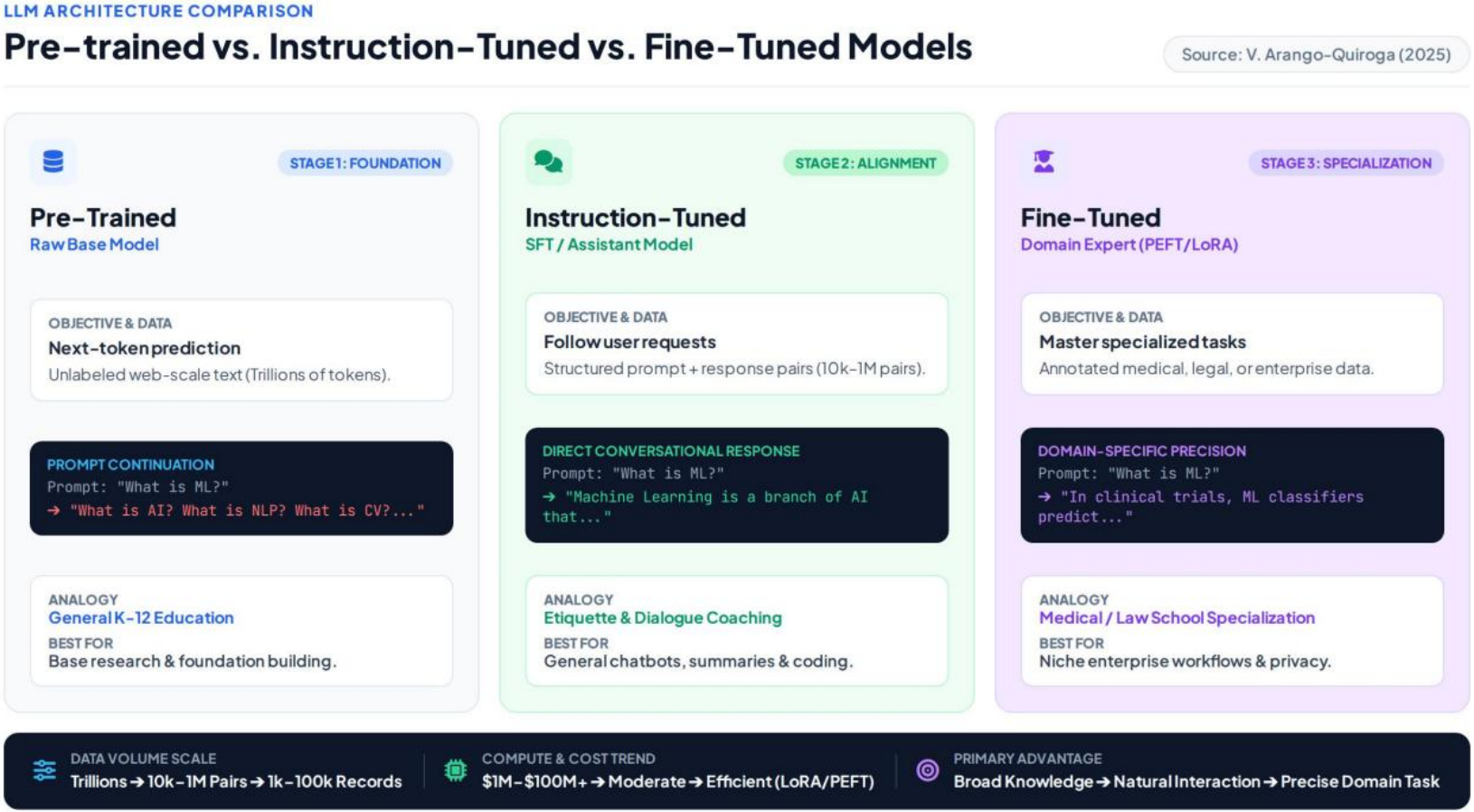
- **Statistical Language Models (1980s–2000s):**
  Used probabilities from large text collections to predict the next word.
  Example: n-gram models.
- **Neural Language Models (2000s–2010s):**
  Used neural networks to understand word relationships and predict words more effectively.
  Example: RNNs and LSTMs.

Humza Naveed, Asad Ullah Khan, Shi Qiu, Muhammad Saqib, Saeed Anwar, Muhammad Usman, Naveed Akhtar, Nick Barnes, Ajmal Mian, “A Comprehensive Overview of Large Language Models”, ACM Transaction on Intelligent System and Technology, vol 16, no 5, p 106, Aug 2025, 72 pages, url https://doi.org/10.1145/3744746.

- **Pre-trained Language Models (PLMs) (2018–):**
  Models were first **pre-trained on huge amounts of text** and then adapted to specific tasks.
  Example: BERT and GPT.
- **Large Language Models (LLMs) (2020s–present):**
  Much larger models trained on massive datasets that can perform many tasks such as **writing, summarizing, translating, answering questions, and coding**.
  Example: GPT, LLaMA, and Gemini.

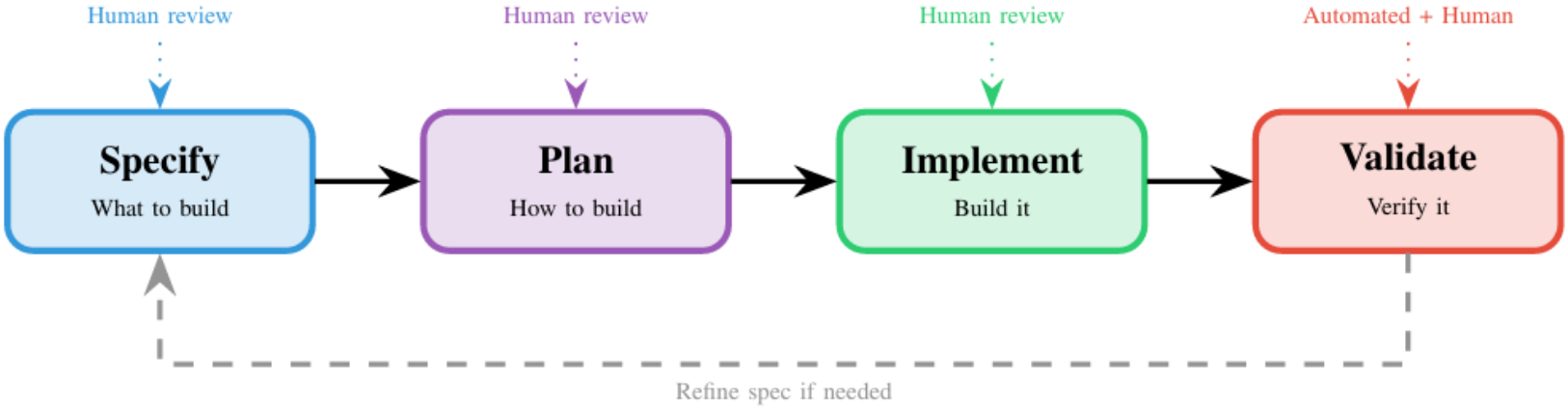


- Chronological display of LLM releases: blue cards represent “pre-trained” models, while orange cards correspond to “instruction-tuned” models.
- Models on the upper half signify open-source availability, whereas those on the bottom are closed-source.



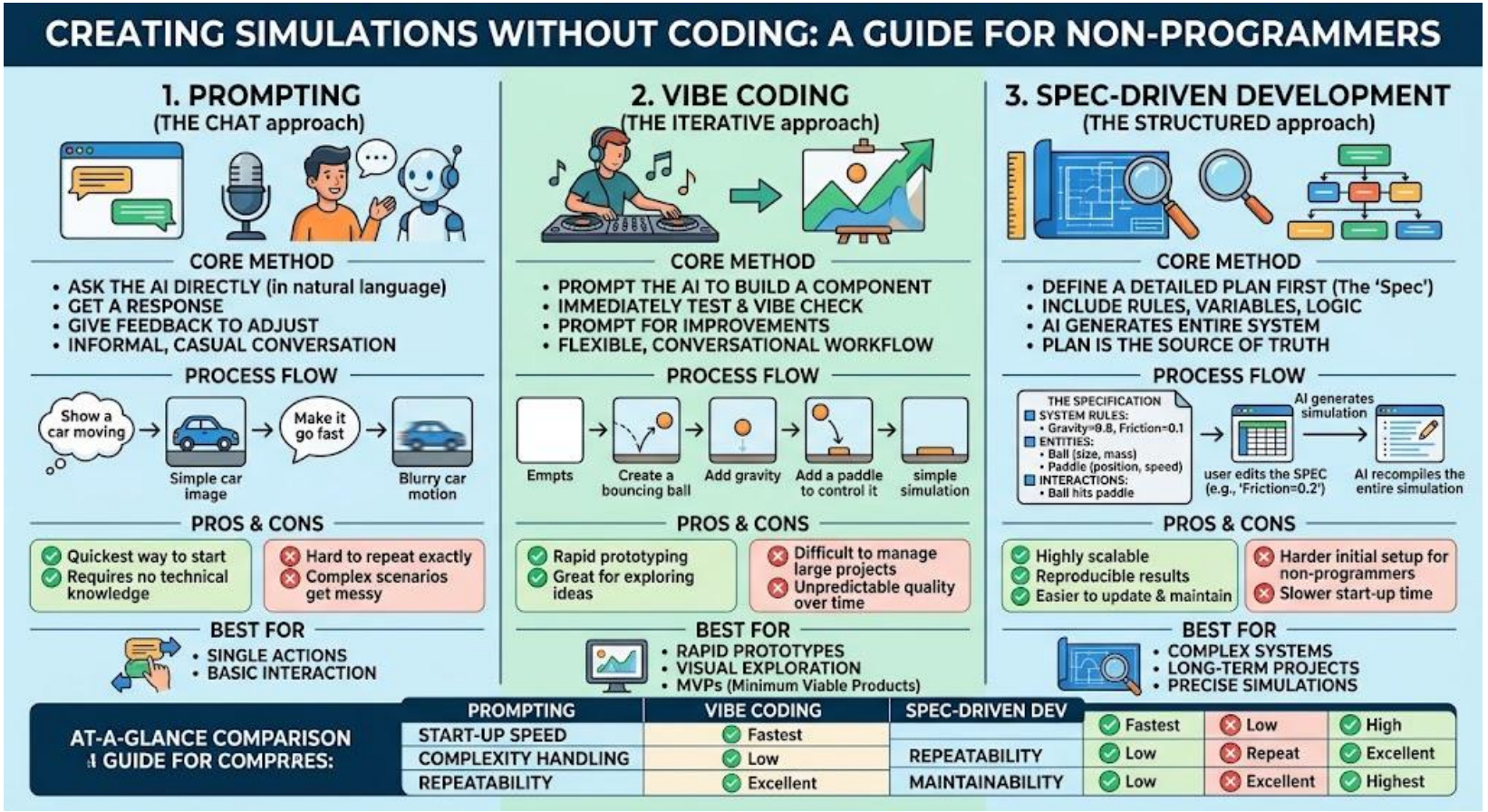
Victor Arango-Quiroga, "Pre-training, Fine-tuning & Instruction Tuning: What's the Difference?", Medium, 9 Mar 2026, url https://medium.com/p/35cf0b0172c3 [20260929].

SDD Workflow



- Spec-driven development (SDD) inverts the traditional work-flow by treating specifications as the source of truth and code as a generated or verified secondary artifac.

Deepak Babu Piskala, "Spec-Driven Development:From Code to Contract in the Age of AI Coding Assistants", arXiv:2602.00180v1 [cs.SE], Fri, 30 Jan 2026 04:45:42 UTC, url https://doi.org/10.48550/arXiv.2602.00180.



Gemini 3.6 Flash, "Comparing AI Simulation Methods", Gemini, 29 Sep 2026, url https://share.gemini.google/2nnPCarnocY4 [20260929].

doppler effect simulation

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Doppler Effect Simulation

- Based on these findings, ChatGPT shows potential as a tool for developing AI-based physics simulations.
- This approach demonstrates that teachers can generate customised simulations for specific learning needs, while enabling stu dents to explore physics ideas more deeply, regardless of their web-programming expertise.
- However, ongoing evaluation remains necessary to ensure that its use meets the demands of effective and contextual physics learning.

Ghonia Millata Mallia Shofa, Tia Jannah Tertia, Sparisoma Viridi, “From prompt to physics: evaluating Doppler effect simulation generated by ChatGPT”, Physics Education, vol 61, no 1, p 015029 (10pp), Feb 2026, url https://doi.org/10.1088/1361-6552/ae272c.

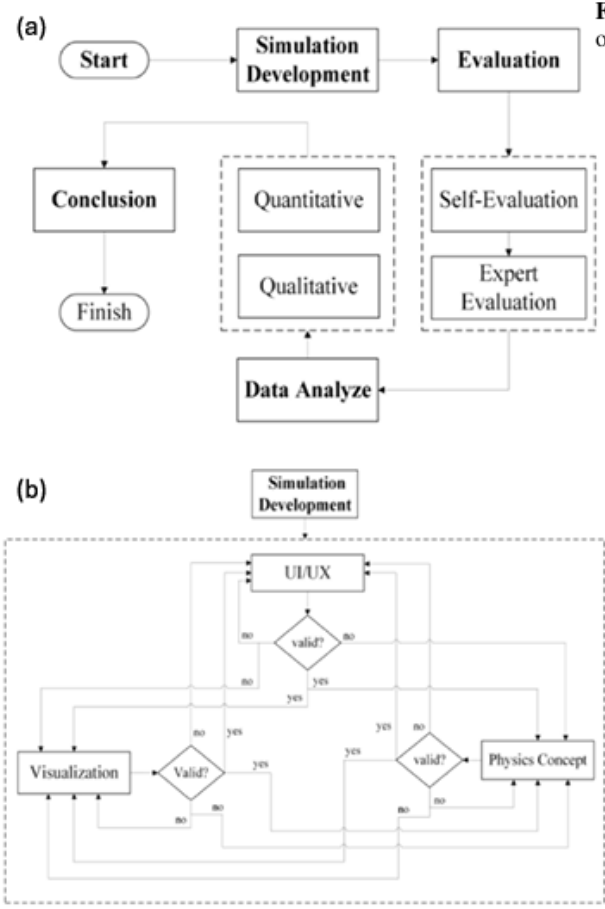


Figure 1. (a) Research flowchart, (b) simulation development flowchart.

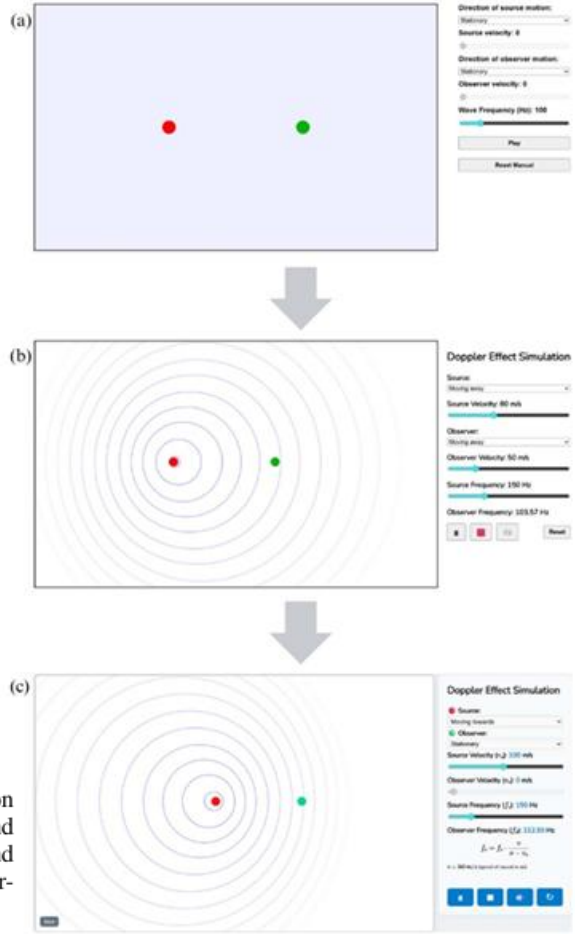


Figure 2. Stages of the Doppler effect simulation development: (a) interface design focusing on UI and UX aspects, (b) addition of physics parameters and visualisation, and (c) final version with improved interface and translated into English.

Table 2. Comparison of the observer's perceived sound frequency obtained from the Doppler effect equation, the simulation display, and the Phyphox measurement.

Case	Source	Observer	v_s (m s^{-1})	v_o (m s^{-1})	f_o (Hz)			Error
					Equation	Simulation	Phyphox	
1	Stationary	Stationary	0	0	200	200	199.96	0.04
2	Stationary	Moving towards	0	50	229.41	229.41	229.34	0.07
3	Stationary	Moving away	0	50	170.59	170.59	170.58	0.01
4	Moving towards	Stationary	50	0	206.06	206.06	205.97	0.09
5	Moving towards	Moving towards	50	50	268.97	268.97	269	0.03
6	Moving towards	Moving away	50	50	200	200	199.96	0.04
7	Moving towards	Moving away	50	80	179.31	179.31	179.29	0.02
8	Moving away	Stationary	50	0	174.36	174.36	174.3	0.06
9	Moving away	Moving towards	50	50	200	200	199.96	0.04
10	Moving away	Moving towards	50	80	215.38	215.38	215.44	0.06
11	Moving away	Moving away	50	50	148.72	148.72	148.58	0.14
Mean error								0.05

Table 4. Evaluators' feedback on the physics concept aspect.

No.	Feedback
1.	The observer's initial position could not be freely determined, so assessments related to the observer's movement towards the source could not be attempted from other positions.
2.	The direction of movement was correct, but the options did not include the direction of movement (right or left), so users had to adjust whether moving towards or moving away meant to the right or left.
3.	The representation of velocity values was correct. The velocity values were well limited to a maximum of 200 m s^{-1} because velocity that were too high for mechanical cases were not physical.

Table 5. Evaluators' Feedback on the user interface (UI) and user experience (UX) aspect.

No.	Feedback
1.	The interface design is quite comfortable but requires a wide screen and is less responsive when the browser screen is narrow.
2.	The slider is easy to use, but an input value box should be provided so that the velocity value is not always an integer. Then, there should be a set number of significant digits because significant digits are essential in physics.
3.	The simulation display is clear and the differences between the elements are visible. However, it would be better if the sound source and observer icons were different. For example, the sound source could be an ambulance and the observer could be a car or motorbike.
4.	I do not think the volume and sound disturb the user. However, when I pressed pause and start repeatedly, the sound was not heard on the umpteenth attempt. After refreshing the simulation page, the sound could be heard again.

electric field visualization

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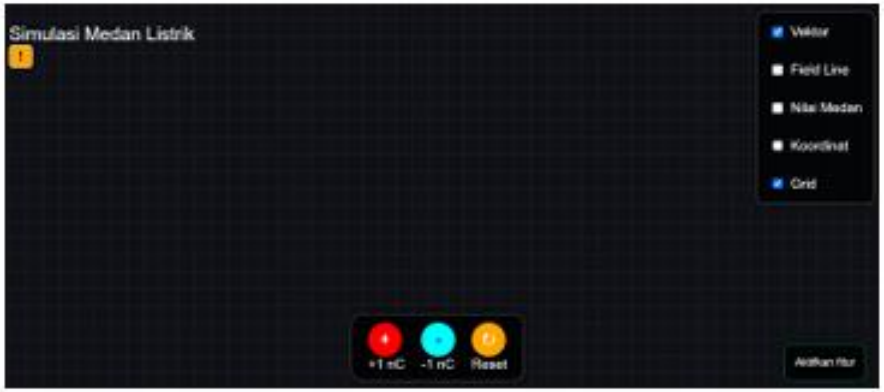
Electric Field Visualization

- This study demonstrates that the LLM-assisted iterative prompt engineering approach can effectively support the development of web-based physics simulations, even for topics with higher conceptual and computational complexity such as electric fields.

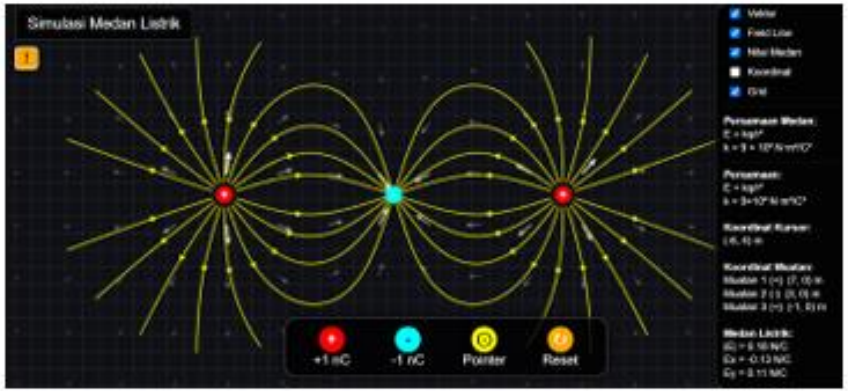
Tia Jannah Tertia, Ghonia Millata Mallia Shofa, Meita Puteri Handayani, Sparisoma Viridi, “Development of ChatGPT-Based Physics Simulations: From the Doppler Effect to Electric Field Exploration”, The 15th International Physics Seminar 2026 (IPS 2026), 21 Jun 2026, Jakarta Indonesia, ABS-78, url https://confgate.net/2026/ips/kfz/abstract/RcNn48P7r [20260927].

- Compared with the Doppler effect simulation used as a benchmark, the electric field simulation introduced greater complexity in physics implementation, dynamic visualisation, and user interaction, yet the approach remained effective in producing an accurate and interactive simulation.
- Expert evaluation confirmed that the developed simulation was appropriate across the physics concept, visualisation, and UI/UX components

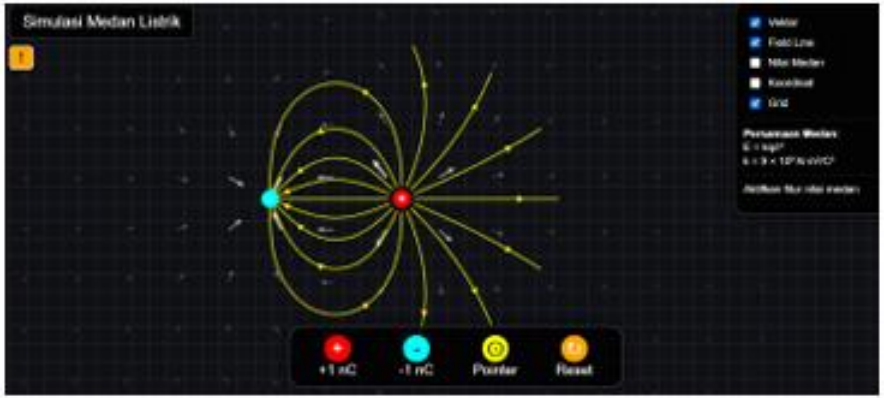
- However, the findings also highlight that human expertise remains essential for validating physics accuracy, ensuring numerical stability, and refining educational usability.



(a)



(c)



(b)

Figure 3. Progressive development of the electric field simulation generated through iterative prompt engineering: (a) initial user interface and control panel, (b) implementation of electric field vectors and field-line visualisation, and (c) final version featuring multi-charge configurations, field superposition, dynamic field-line generation, and interactive information displays.

Table 2. Expert Evaluation Results of the Developed Electric Field Simulation

Evaluation components	Indicators	Score (max.15)	Percentage (%)	Average Percentage (%)
Physics Concept	Accuracy of the Inverse-Square Law	15	100	97,78
	Vector Representation	15	100	
	Validity of the Superposition Principle	14	93,33	
Visualisation	Clarity of Field-Line Visualisation	15	100	100
	Real-Time Dynamics	15	100	
UI/UX	Control Responsiveness	14	93,33	93,33
	Information Readability	14	93,33	
	Navigation and Intuitiveness	14	93,33	

futsal goal probability simulation

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Futsal Goal Probability Simulation

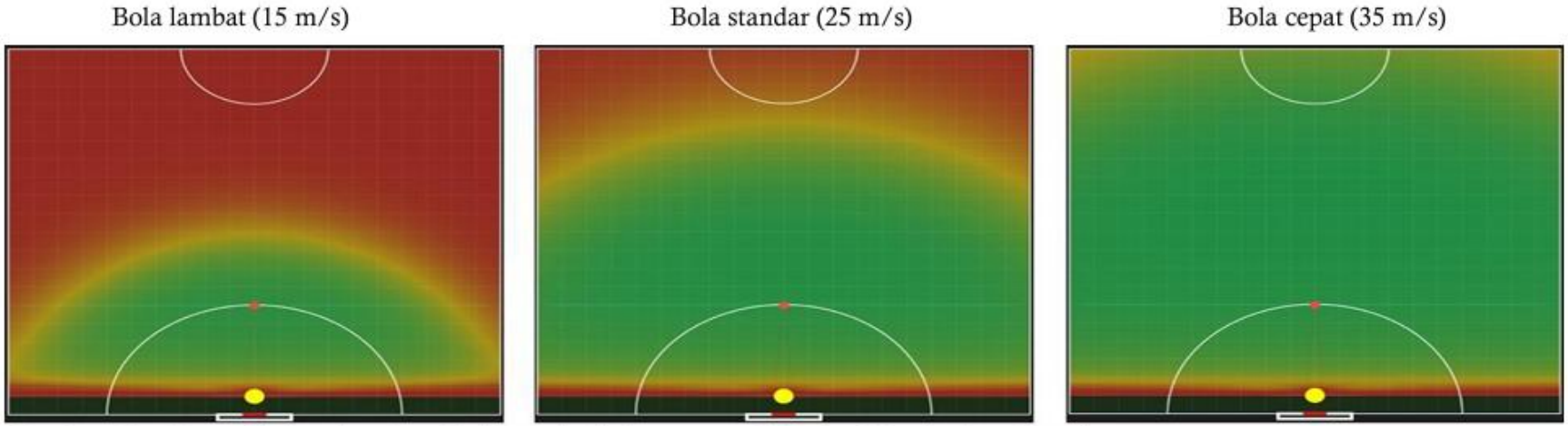
- This study shows that goal-scoring opportunities in one-on-one futsal situations result from the simultaneous interaction of geometric and kinematic factors.
- Higher ball speed significantly expands the scoring zone due to human motor-response limitations.
- The developed Agent-Based Model mechanistically explains goal-scoring opportunities and has potential as a physics-based analytical framework for sports science.

Anzet Dianto Pebriwan, Mery Cintia Afrilya Sitinjak, Sparisoma Viridi, Asril Pramutadi Andi Mustari, “Pemodelan Peluang Gol pada Futsal Satu Lawan Satu Menggunakan Simulasi Kinematika dan Geometri Shooting”, National Conference on Footbal and Science, 5-6 Sep 2026, Bandung, Indonesia, url https://ncfs-indonesia.com/call-for-papers/ [20260927]

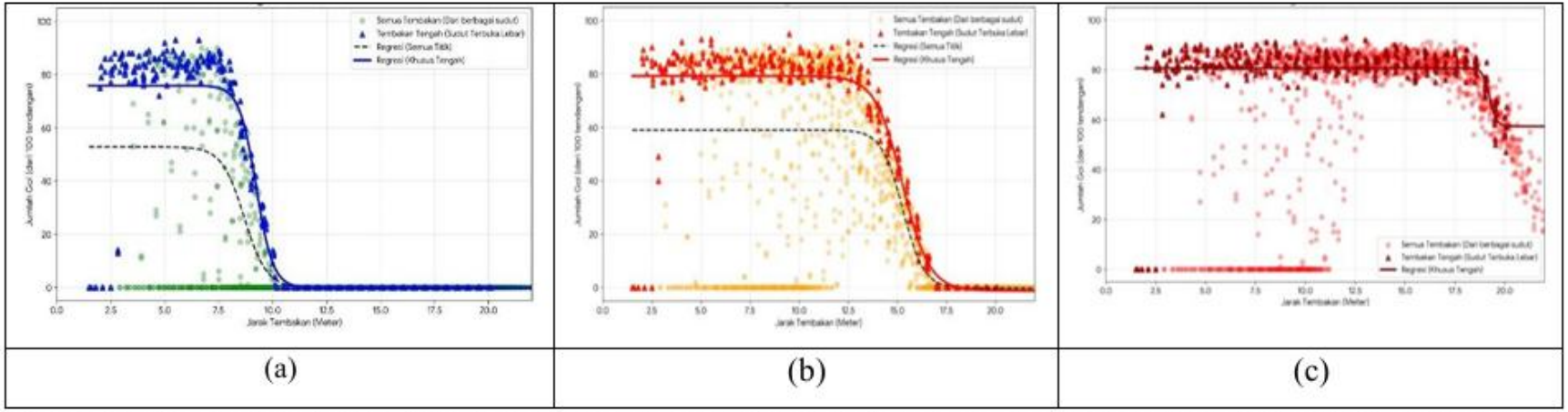
- Based on the inflection point of the sigmoid curve, the critical distance (x_0) at which goal probability reaches 50% increases from 9.23 m at 15 m/s to 15.26 m at 25 m/s and 19.24 m at 35 m/s. This indicates that high-speed shots can substantially reduce the goalkeeper's available reaction time, diminishing the effectiveness of distance as a defensive parameter.

- The angular-opening threshold required to reach a 50% goal probability decreases from 0.21 rad at 15 m/s to 0.09 rad at 35 m/s. Thus, higher ball speed significantly expands the effective shooting area. The critical distance (d_0) at which goal probability reaches 50% shifts from 5.78 m at 15 m/s to 13.34 m at 35 m/s, indicating that increased kinetic energy can more than double the attacker's "danger zone."

- A limitation of this study is that the model does not yet incorporate three-dimensional ball trajectories or variations in the goalkeeper’s adaptive decision-making.



Gambar 2. xG teoritis berdasarkan kelajuan bola berurutan 15 m/s, 25 m/s dan 35 m/s.



Gambar 3. Kurva regresi sigmoid hubungan jarak tembak terhadap frekuensi gol pada kelajuan (a) 15 m/s; (b) 25 m/s ; (c) 35 m/s

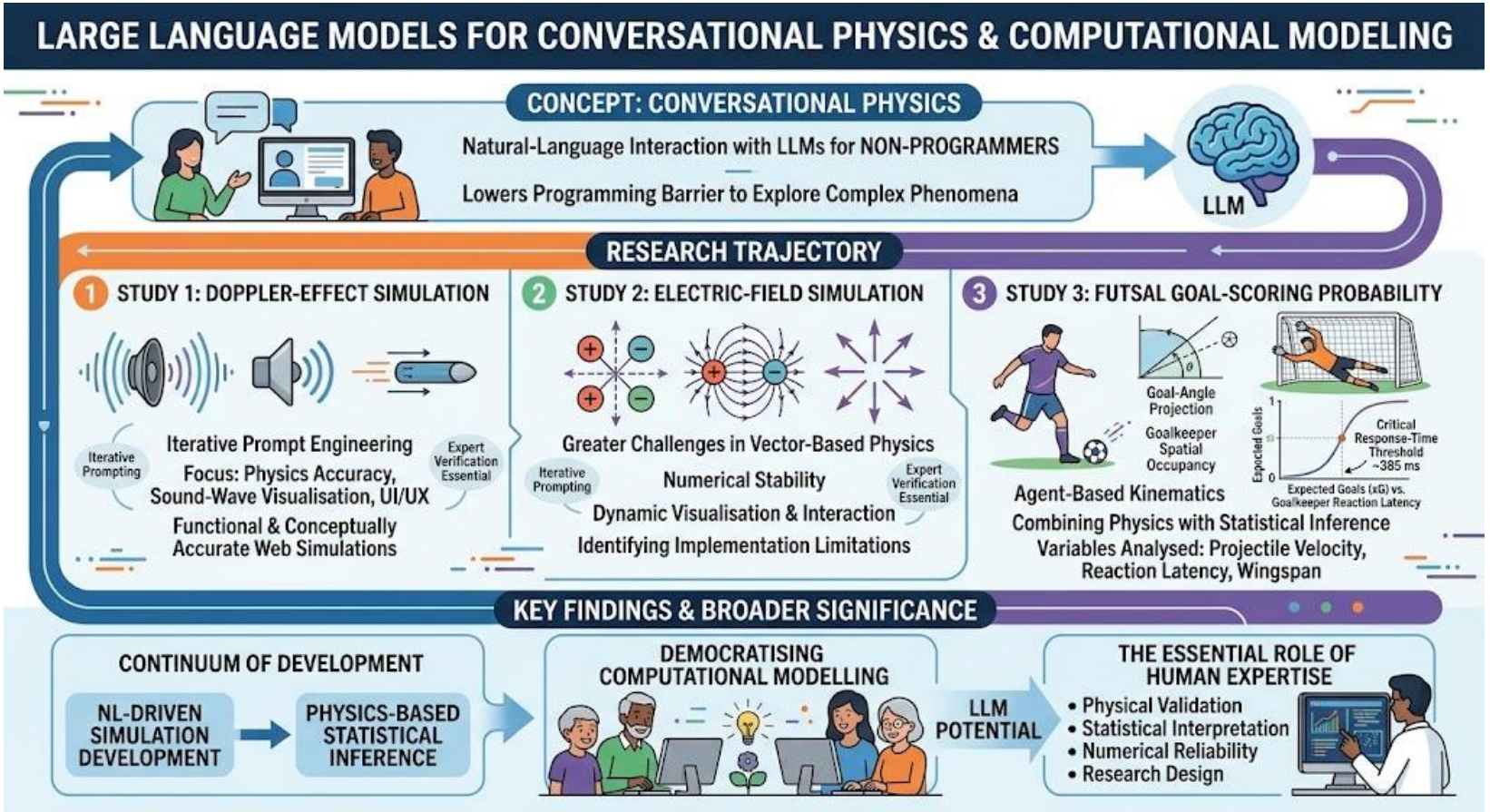
closing

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Summaries

- These studies illustrate a continuum from natural-language-driven simulation development to physics-based statistical inference.
- The keynote highlights the potential of LLMs to democratise computational modelling while emphasising the continuing role of human expertise in physical validation, statistical interpretation, numerical reliability, and research design.

- The broader significance lies in establishing conversational interfaces as a bridge between domain knowledge, simulation, and statistical analysis for researchers, educators, and non-programmers.



Gemini 3.6 Flash, "Conversational Physics Infographic Design", Gemini, 29 Sep 2026, url https://share.gemini.google/nKm1Yjaqk7jl [20260929].

Discussion

- Please give us suggestions and comments for improvement.

Thank you

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